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percolation of this hardware-in-loop testing into the RF system design area for validation," explains Satish Mohanram, technical marketing manager, National Instruments-India.

"In telecom field simulation is a very important tool to test network elements, handsets, etc. The cost of a real network is much more than that of the simulator. First-stage design, testing of network instruments and mobile handsets, and application tests are performed using a simulated environment. Simulations also help to achieve repeatable, error-free testing. This reduces dependency on the network, its access, error and testing times," explains Madhukar Tripathi, manager-telecom segment and indirect channel, Anritsu India.

"Many test houses use simulation to verify RF, protocol and particular applications on mobile handsets. Network performance too can be tested using a simulator to see how a base transceiver station (BTS) will behave when hundreds or thousands of calls are made within the same BTS area, how download of music will be affected when a user moves from GSM to WCDMA area (handover situation), or how bandwidth is utilised when many users are downloading particular applications."

In India, even when the LTE network was not ready, test houses and labs were able to test devices using simulators.

All is not always well

While there are advantages of simulation, design firms also face some disadvantages during testing. Simulations are usually based upon laws and theories, which may not be true in re-

ality. So actual results often do not get simulated. Additionally, one tiny error in the input of parameter setting can jeopardise the whole setup and results.

Anand shares an incident: "We at Knewron once faced near-failure situation due to simulation gaps in one of the critical projects. The simulation tool would show that the design is perfect while in reality there were stability and reliability issues; and these issues we could simulate only in the user environment on-site. When we realised about these gaps, we immediately switched to real-time testing and got the project up and running quickly."

"The point I am making here is that simulations may not give near-reality results always as the user environment can play a crucial role in product usage and performance. Design firms need to keep tab on such critical points all the time."

Design firms await simulation tools that learn from past simulations and actual results. Self-learning capability may help these tools to shrink the gap between simulation results and reality.

Empowering the design engineer

Engineering decisions, which were the domain of analysts till now, are coming to the desk of design engineers. Gone are the days when analysis or simulation used to be a post-design activity. Today, before taking any decision about design, engineers have to simulate and validate it. The simulation domain is growing fast, keeping up with the rapid technological advances that we are witnessing in this era! ●

The author is a tech correspondent at EPI

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